

Phosphorylated L-Serine Biosynthesis: A Reusable Three-Reaction Module

A commissioned review-style synthesis of the SerA–SerC–SerB pathway that converts the glycolytic intermediate 3-phospho-D-glycerate to L-serine. Search date: 2026-07-25. Citations given as PubMed IDs (PMID).

1. Executive Summary

Phosphorylated L-serine biosynthesis is a compact, ancient, and near-universal metabolic module that diverts a fraction of glycolytic flux toward the amino acid L-serine. It comprises exactly three enzymatic reaction roles, executed in an obligatory order: (1) **SerA / phosphoglycerate dehydrogenase (PGDH/PHGDH)** oxidizes 3-phospho-D-glycerate to 3-phosphohydroxypyruvate using NAD⁺; (2) **SerC / phosphoserine aminotransferase (PSAT/PSAT1)** transaminates that keto-acid to O-phospho-L-serine using L-glutamate and the cofactor pyridoxal-5'-phosphate (PLP); and (3) **SerB / phosphoserine phosphatase (PSP/PSPH)** hydrolyzes the phosphate ester to release free L-serine, using Mg²⁺ and a covalent phospho-aspartyl intermediate. In all non-photosynthetic organisms, including mammals, this "phosphorylated pathway" is the major route of *de novo* serine synthesis ([PMID: 30619878](#)).

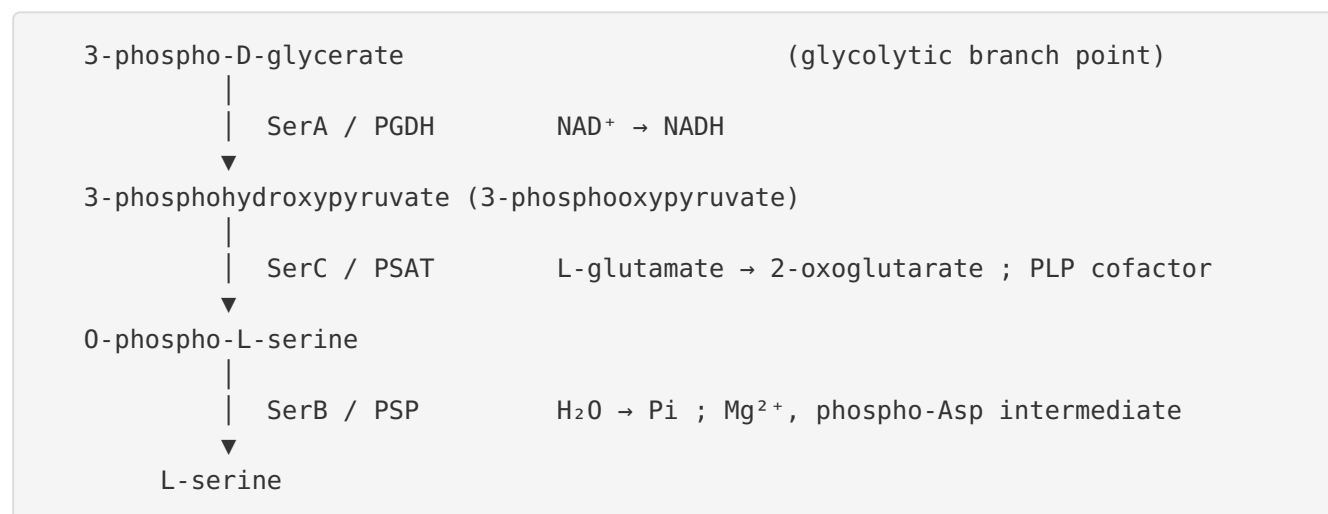
The most important conceptual point for reviewers is that the pathway is best defined by **reaction roles**, not by protein identity or genomic arrangement. The same three chemistries are carried out by proteins drawn from three independently ancient superfamilies — the D-isomer-specific 2-hydroxyacid dehydrogenases (SerA), the fold-type I PLP aminotransferases (SerC), and the haloacid dehalogenase (HAD) hydrolases (SerB). Several of these proteins also perform chemically distinct "moonlighting" reactions (SerA producing the oncometabolite D-2-hydroxyglutarate; SerC transaminating a vitamin-B6 precursor) that share the enzyme but lie **outside** the pathway boundary. Defining the module by its serine-forming reaction roles cleanly excludes these side activities and the parallel plant photorespiratory/glycerate routes that reach the same product by different means.

The system's core chemistry is deeply conserved, but its **regulation is lineage-variable**. Bacteria and plants throttle flux at SerA through V-type allosteric feedback: L-serine binds accessory ACT (and, in some actinobacteria, ASB) domains to shift the enzyme population toward an inactive conformation. Mammals rely instead on expression level, post-translational modification, and gene copy number, which is why PHGDH behaves as a rate-limiting oncology node. Loss-of-function in humans produces a graded serine-deficiency disease spectrum from lethal Neu-Laxova syndrome to childhood intellectual disability, much of it treatable with L-serine supplementation. Downstream, the L-serine product is the obligate precursor for glycine, one-carbon units, and — in the brain — the NMDA-receptor co-agonist D-serine.

2. Definition and Biological Boundaries

What is included

The biological system under review is a **three-reaction pathway** with the following stoichiometric backbone:



The module "begins" at the withdrawal of 3-phospho-D-glycerate from glycolysis and "ends" at free L-serine. It is defined independently of genomic arrangement: the three genes may be scattered, clustered, or fused across genomes, but the reusable reaction roles remain the same. The pathway is the major *de novo* source of L-serine "in all non-photosynthetic organisms, including mammals" (PMID: 30619878).

What is adjacent but should be treated separately

- **Downstream serine fates.** Serine hydroxymethyltransferase (SHMT) converts L-serine to glycine and a one-carbon unit; serine racemase converts L-serine to D-serine. These are consumers of the module's product, not steps within it.
- **Plant alternative routes.** In plants, two additional pathways make serine: the **photorespiration-associated glycolate pathway (GPSB)** and the **glycerate pathway**. These are genetically and functionally distinct from the **phosphorylated pathway (PPSB)** (PMID: 37804096). They reach the same product by different molecular means and in different compartments/tissues.
- **Boundary-excluded moonlighting.** Some SerA proteins reduce α -ketoglutarate to D-2-hydroxyglutarate; some SerC proteins transaminate a vitamin-B6 (PLP) biosynthetic precursor. These are the **same proteins** doing **different chemistry** and are explicitly outside the pathway boundary.

Competing definitions

The main definitional tension in the literature is **protein-identity vs. reaction-role**. Because individual enzymes moonlight and because genomic arrangements differ, a protein-centric definition either over-includes (folding in 2-HG or B6 chemistry) or fragments across genomes. The reaction-role definition adopted here is more robust and is the one that best matches the working scope.

3. Mechanistic Overview

Step 1 — SerA / PGDH: NAD⁺-dependent oxidation

PGDH catalyzes the first, committed oxidation of 3-phospho-D-glycerate to 3-phosphohydroxypyruvate with reduction of NAD⁺. Classic transient-kinetic work established an ordered ternary-complex mechanism with hydride transfer flanked by isomerization steps, and identified serine as an allosteric inhibitor that locks the enzyme in a conformation incapable of catalysis — the defining feature of a **V-type** allosteric enzyme (PMID: 14154). In mammals this is the **rate-limiting** step of *de novo* serine synthesis, "facilitating the conversion of 3-phosphoglycerate to 3-phosphohydroxypyruvate" (PMID: 38945960).

Step 2 — SerC / PSAT: PLP-dependent transamination

PSAT is a fold-type I PLP-dependent aminotransferase of subgroup IV (PMID: 15103612). It transfers the amino group from L-glutamate to 3-phosphohydroxypyruvate, yielding O-phospho-L-serine and regenerating 2-oxoglutarate. The reaction runs through the canonical ping-pong PLP mechanism (internal aldimine → external aldimine → ketimine → quinonoid), producing the phosphorylated amino acid and coupling serine synthesis to the glutamate/2-oxoglutarate nitrogen pool.

Step 3 — SerB / PSP: Mg²⁺-dependent, irreversible dephosphorylation

PSP is a Mg²⁺-dependent member of the HAD superfamily — "PSPs and P-type ATPases are members of the haloacid dehalogenase (HAD)-like hydrolase family" (PMID: 11342136). It hydrolyzes O-phospho-L-serine to L-serine and inorganic phosphate via a **covalent phospho-aspartyl enzyme intermediate**, being "a member of a large class of enzymes that catalyze phosphoester hydrolysis using a phosphoaspartate-enzyme intermediate" (PMID: 12051918). Crystallographic "snapshots" of the *Methanococcus jannaschii* enzyme captured the full reaction cycle (open, substrate-bound, AlF₃ transition-state mimic, BeF₃[−]/phospho-enzyme, product), making this one of the best-resolved phosphatase mechanisms known. Because dephosphorylation is thermodynamically favorable and effectively irreversible, this step commits flux and makes the pathway unidirectional.

Obligatory / conditional / accessory classification

Step	Enzyme	Chemistry	Status
1	SerA / PGDH	NAD ⁺ oxidation	Obligatory , rate-limiting (mammals), feedback-controlled (bacteria/plants)
2	SerC / PSAT	PLP transamination	Obligatory
3	SerB / PSP	HAD dephosphorylation	Obligatory , irreversibility-conferring
—	Accessory domains (ACT/ASB on SerA)	Effector sensing	Conditional (present in some lineages)
—	SerA→2-HG; SerC→B6 precursor	Moonlighting	Accessory / outside boundary

4. Major Molecular Players and Active Assemblies

SerA / PGDH — a V-type allosteric dehydrogenase in three structural classes

PGDHs "exist with at least three different structural motifs and the enzymes from different species display distinctly different mechanisms" (PMID: 22023909). These motifs differ in accessory regulatory domains: catalytic domain alone; catalytic + ACT; catalytic + ACT + ASB. In *E. coli*, "the catalytic activity is regulated allosterically through binding of l-serine to a distinct structural domain, termed the ACT domain" (PMID: 22023909). The mechanistic basis of this V-type regulation is "a reduction in the population of active species rather than a differential decrease in the velocity of active species" (PMID: 19924905); serine inhibition is positively cooperative with a Hill coefficient of ~2 (PMID: 12183470).

In *Mycobacterium tuberculosis*, an additional **ASB domain** and an **anion-binding site** couple substrate and effector for synergistic regulation, and NADH can compete with substrate at the allosteric site to relieve substrate inhibition (PMID: 19388702). Notably, only certain pathogenic mycobacteria carry the full complement of basic residues needed for high serine sensitivity, so serine feedback is not universal even within a single family (PMID: 25698123).

SerC / PSAT — fold-type I PLP aminotransferase, subgroup IV

SerC belongs to "the fold-type I pyridoxal 5'-phosphate (PLP)-dependent enzymes," and *E. coli* phosphoserine aminotransferase specifically "belongs to subgroup IV" (PMID: 15103612). It functions as a homodimer with two shared active sites at the subunit interface — the general architecture of aspartate-aminotransferase-like enzymes.

SerB / PSP — HAD-superfamily phosphotransferase

SerB is a Mg²⁺-dependent HAD hydrolase (PMID: 11342136) that uses a phospho-aspartate covalent intermediate (PMID: 12051918). Structural work extends beyond archaea to bacterial pathogens: the *Brucella melitensis* PSP structure and its Mg²⁺-substitution studies confirm the conserved catalytic scaffold and its druggability (PMID: 40719280). The HAD framework is broad — related phosphatases such as PHOSPHO1 were modeled directly on the *M. jannaschii* PSP structure (PMID: 14983068) — underscoring that PSP substrate specificity is conferred by a small set of active-site residues rather than the fold itself.

Downstream partners (outside the module)

L-serine is handed off to SHMT for glycine/one-carbon metabolism ([PMID: 42015255](#)) and, in the brain, to serine racemase for D-serine production ([PMID: 41735678](#)).

5. Evolutionary and Cell-Biological Variation

Deep conservation of chemistry, divergence of regulation

The three catalytic scaffolds derive from **three independently ancient superfamilies**, implying the module was assembled by recruiting pre-existing enzyme folds rather than co-evolving from a single ancestor. The chemistry is essentially invariant across bacteria, archaea, plants, and mammals; the **regulatory apparatus is what varies**.

Lineage	Primary flux control	Structural / molecular basis
Bacteria (<i>E. coli</i>)	L-serine V-type feedback	ACT domain on SerA (PMID: 22023909)
Pathogenic actinobacteria (<i>M. tuberculosis</i>)	Synergistic serine + anion feedback	ACT + ASB domains, anion-binding site (PMID: 19388702)
Plants (<i>Arabidopsis</i>)	Compartment/tissue specialization	Distinct gene sets: PPSB vs GPSB vs glycerate (PMID: 37804096)
Mammals	Expression, PTM, gene amplification, allostery	Transcriptional/oncogenic control (PMID: 38945960)

Plants: compartment- and tissue-specialized routes

Plants uniquely run three parallel serine routes. The literature explicitly contrasts "the phosphorylated pathway of Ser biosynthesis (PPSB) with the photorespiration-associated glycolate pathway of Ser biosynthesis (GPSB)" ([PMID: 37804096](#)). The GPSB is restricted to photosynthetic cells; the PPSB is essential in **heterotrophic tissues** where photorespiration cannot supply serine. Genetic dissection demonstrates "the essential role of PSP1 for embryo and pollen development, and for proper root growth" ([PMID: 24299976](#)), and the PGDH isoform

EDA9 is likewise essential for embryo and male gametophyte development (PMID: 24304635). This is the clearest example of the same reaction module being deployed in a cell-type-specific, spatially non-overlapping manner alongside alternative routes.

Mammals: PHGDH as a rate-limiting oncology node

In mammals PHGDH "functions as the primary rate-limiting enzyme in the serine biosynthesis pathway" (PMID: 38945960), and elevated PHGDH activity in cancers "is mediated through genetic amplification, posttranslational modification, increased transcription, and allosteric regulation" (PMID: 38945960). PHGDH supports one-carbon units, antioxidant capacity, and epigenetic methyl-donor supply across breast cancer, melanoma, glioma, and pancreatic cancer. Structure-based inhibitors have been developed (PMID: 36594670; PMID: 42263187), but a documented escape route is that "cancer cells can activate alternative metabolic pathways, such as glutaminolysis, to evade PHGDH inhibition" (PMID: 40843962). In PDAC, PHGDH loss unexpectedly suppresses mTOR signaling and glutamine uptake rather than forcing serine auxotrophy (PMID: 41959234). This illustrates that the pathway's "essentiality" is context-dependent, not absolute.

Downstream cell-biology: the astrocyte–neuron serine shuttle

In brain, "l-Serine is synthesized from glucose in astrocytes, exported to neurons, and converted by serine racemase to d-serine, an NMDAR co-agonist" (PMID: 41101375); D-serine is "an endogenous D-amino acid and a crucial co-agonist for the N-methyl-D-aspartate (NMDA) receptor" (PMID: 41735678). Dysregulated D-serine homeostasis is implicated in schizophrenia, Alzheimer's disease, and injury-associated synaptic pruning — biology that depends on, but is distinct from, the phosphorylated module itself.

6. Constraints, Dependencies, and Failure Modes

Ordering constraints

The three steps must occur in sequence because each produces the obligate substrate of the next: SerA supplies 3-phosphohydroxypyruvate to SerC, and SerC supplies O-phospho-L-serine to SerB. No physiologically significant shortcut bypasses the phosphorylated intermediates within this module. The **SerB dephosphorylation is irreversible**, so flux, once committed past step 3, cannot run backward — a thermodynamic ratchet that also prevents futile cycling.

Cofactor and compartment dependencies

- SerA requires **NAD⁺/NADH**; its direction is sensitive to the cellular redox couple.
- SerC requires **PLP** (vitamin B6) and a supply of **L-glutamate** as amino donor, coupling serine synthesis to nitrogen metabolism.
- SerB requires **Mg²⁺**; substituting the metal alters activity ([PMID: 40719280](#)).
- The mammalian module is cytosolic; the plant module is compartment/tissue-restricted (PPSB in heterotrophic tissue vs. GPSB in photosynthetic cells).

Failure modes

Recessive loss-of-function in **PHGDH**, **PSAT1**, or **PSPH** causes human serine-biosynthesis defects that "have a broad phenotypic spectrum ranging from Neu-Laxova syndrome, a lethal multiple congenital anomaly disease at the severe end to a childhood disease with intellectual disability at the mild end" ([PMID: 26960553](#)). Many patients respond to L-serine supplementation, especially when started early — demonstrating that the deficit is one of biosynthetic *supply*, correctable by bypassing the pathway with dietary serine. Large-scale variant-effect mapping for PSAT1 now allows functional classification of nearly all single-nucleotide variants and predicts biallelic (compound-heterozygous) genotype severity ([PMID: 37812589](#)).

Evidence ruling out alternative paths

- The plant genetic dissection separating PPCB from GPSB/glycerate shows that these are **not interchangeable** in vivo: PPCB knockouts fail in heterotrophic tissue despite an intact photorespiratory route ([PMID: 37804096](#); [PMID: 24299976](#)).
- The V-type mechanism of SerA feedback (population shift, not velocity change) is directly established by transient kinetics, ruling out simple competitive-inhibition models ([PMID: 19924905](#)).

7. Controversies and Open Questions

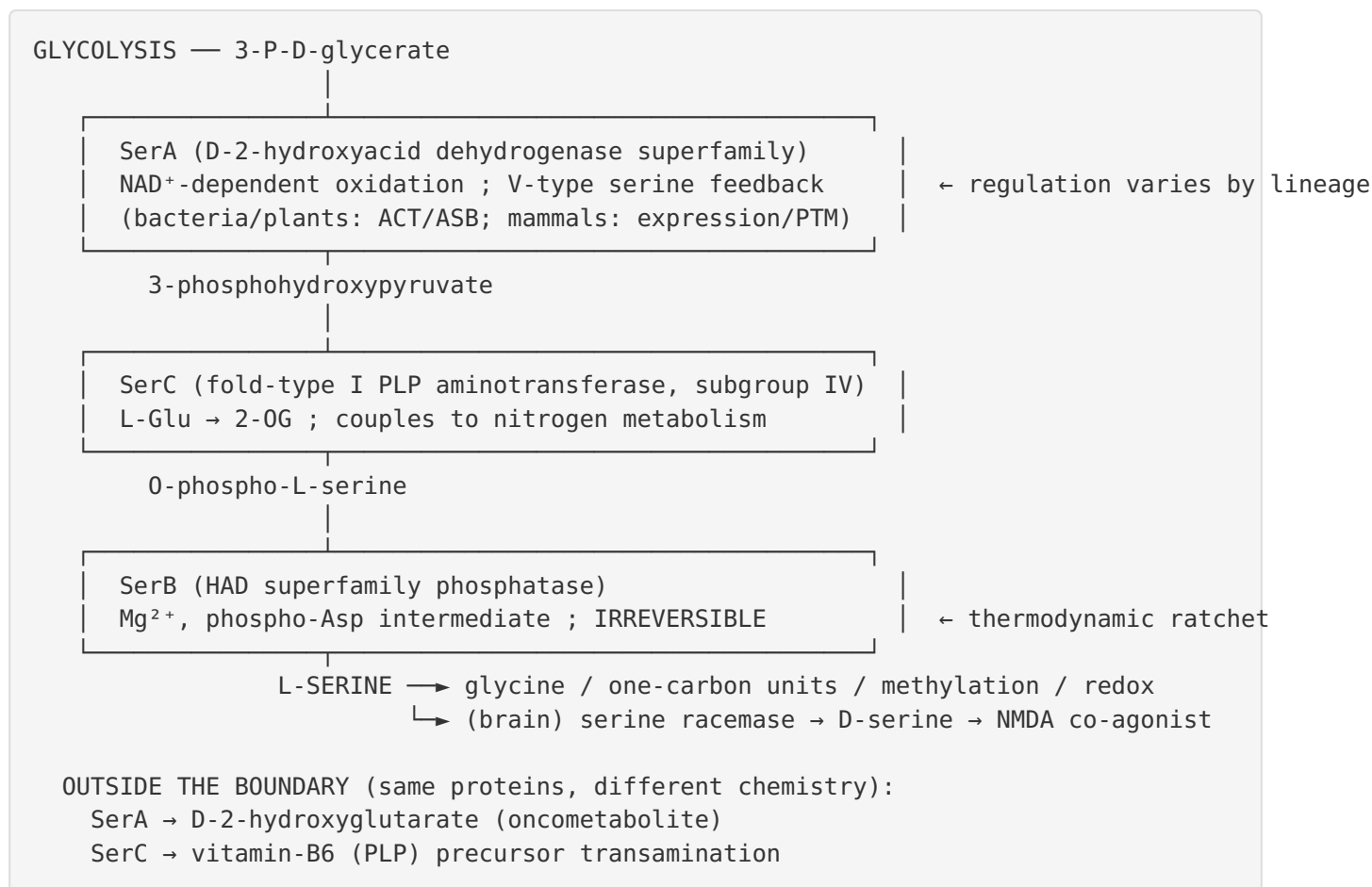
1. Is **PHGDH** essential in cancer because of serine, or because of moonlighting? Human PHGDH, "in addition to catalyzing oxidation of 3-phosphoglycerate, ... catalyzes NADH-dependent reduction of α -ketoglutarate (AKG) to the oncometabolite d-2-hydroxyglutarate (d-2HG)" ([PMID: 25406093](#)), and "knockdown of PHGDH is not fully rescued by exogenous

serine, suggesting possible additional growth-promoting roles for the enzyme" (PMID: 25406093). Non-metabolic scaffolding functions are also emerging — e.g., PHGDH binding and inhibiting the kinase TAK1 in macrophages (PMID: 40889684). Disentangling serine-dependent from serine-independent contributions is an active controversy with direct implications for inhibitor design.

2. **Which SerA family member best represents the ancestral role?** PGDH exists in ≥ 3 structural classes (PMID: 22023909). The catalytic-domain-only enzymes plausibly reflect the ancestral minimal chemistry, with ACT and ASB regulatory domains as later, lineage-specific elaborations; but this ordering is inferred rather than experimentally reconstructed.
3. **How comparable are cross-organism claims?** Much mechanistic detail comes from *E. coli*, *M. tuberculosis*, and archaeal PSP, whereas disease and cancer data come from human cells. Regulatory conclusions do not transfer directly — e.g., mammals lack the ACT-domain V-type control that dominates bacteria. Reviewers should be wary of extrapolating regulatory logic across lineages.
4. **What sets steady-state serine in specific tissues?** The relative contributions of *de novo* synthesis vs. uptake vs. downstream consumption (SHMT, racemase) differ by tissue and physiological state, and remain incompletely quantified, especially in brain where the astrocyte–neuron shuttle governs D-serine availability (PMID: 41101375).

8. Mechanistic Model / Synthesis

Integrating all eight confirmed findings, the phosphorylated pathway is best modeled as an **ordered, irreversible, reaction-role module with lineage-variable regulation**:



Seven integrating conclusions: (1) three obligatory, ordered steps with SerB making flux unidirectional; (2) three enzymes from three independently ancient superfamilies; (3) regulation differs by lineage (bacterial/plant V-type ACT/ASB feedback vs. mammalian expression/PTM/amplification); (4) plant photorespiratory and glycerate routes reach the same product by different, compartment-restricted means; (5) loss causes a graded human disease spectrum; (6) the product L-serine feeds one-carbon/glycine/D-serine biology; (7) boundary-excluded moonlighting activities justify a reaction-role rather than protein-identity definition.

9. Evidence Base

PMID	Contribution	Role in this review
30619878	Defines the three-enzyme pathway and its status as the major <i>de novo</i> serine source	Foundation (scope, F001)
22023909	Three PGDH structural motifs; ACT-domain serine feedback	SerA classes/ regulation (F002)
19924905	V-type mechanism = reduced active-species population	Feedback mechanism (F002)
12183470	Hill coefficient ~2 cooperativity; NADH-modulated effector binding	Cooperativity (F002)
19388702	ASB / anion-binding site in <i>M. tuberculosis</i> PGDH	Lineage variation
25698123	Serine sensitivity confined to specific pathogenic mycobacteria	Regulatory diversity
15103612	PSAT as fold-type I PLP aminotransferase, subgroup IV	SerC scaffold (F003)
11342136	PSP as HAD-superfamily member (archaeal structure)	SerB scaffold (F003)
12051918	Phospho-aspartyl intermediate; reaction-cycle snapshots	SerB mechanism (F003)
40719280	<i>Brucella</i> PSP structure and Mg ²⁺ dependence	SerB druggability
14983068	PHOSPHO1 modeled on PSP; specificity from few residues	HAD breadth
26960553	Human disease spectrum from PGDH/PSAT/PSP deficiency	Failure modes (F004)
37812589	Large-scale PSAT1 variant-effect map	Variant interpretation
37804096	PPSB vs. GPSB distinction in plants	Plant boundaries (F004)
24299976	PSP1 essential for embryo/pollen/root	Plant essentiality (F004)
24304635	EDA9 PGDH isoform essential for development	Plant variation
38945960	PHGDH rate-limiting; multiple upregulation modes in cancer	Oncology node (F005)

PMID	Contribution	Role in this review
40843962	Glutaminolysis escape from PHGDH inhibition	Resistance (F005)
41959234	PHGDH loss suppresses mTOR/glutamine uptake in PDAC	Context-dependence
25406093	PHGDH produces D-2-hydroxyglutarate; serine-independent role	Moonlighting (F007)
40889684	PHGDH-TAK1 non-metabolic scaffolding	Moonlighting controversy
41735678	D-serine as NMDA co-agonist	Downstream (F006)
41101375	Astrocyte→neuron serine shuttle, racemase step	Downstream boundary (F006)
42015255	SHMT one-carbon consumption of serine	Downstream partner
14154	Original V-type kinetic characterization of PGDH	SerA mechanism
36594670 / 42263187	Structure-based PHGDH inhibitors	Therapeutics

10. Limitations and Knowledge Gaps

- **Organism heterogeneity.** Mechanistic detail derives largely from bacterial/archaeal enzymes, while disease and cancer data derive from human cells. Regulatory conclusions (V-type ACT feedback) are not universal and should not be extrapolated across lineages.
- **Moonlighting attribution.** The quantitative split between PHGDH's serine-forming role and its 2-HG / scaffolding roles in driving cancer phenotypes remains unresolved ([PMID: 25406093](#); [PMID: 40889684](#)).
- **Ancestral reconstruction.** The proposed ordering (catalytic-only ancestor → ACT/ASB elaborations) is inferred from structural comparison, not from experimental ancestral-sequence resurrection.
- **Tissue-level flux.** The contribution of *de novo* synthesis vs. serine uptake across mammalian tissues and physiological states is incompletely quantified.

- **Literature-based synthesis.** This review rests on the cited experimental studies rather than new primary data or statistical analysis.

11. Proposed Follow-up Experiments / Actions

1. **Separation-of-function PHGDH mutants.** Engineer variants that retain 3-PG oxidation but abolish α -KG $\rightarrow$ 2-HG reduction (and vice versa) to quantify each activity's contribution to tumor growth, resolving the central moonlighting controversy.
 2. **Ancestral sequence reconstruction of SerA.** Resurrect and characterize inferred ancestral PGDHs to test whether catalytic-domain-only enzymes represent the ancestral, unregulated state and when ACT/ASB regulation arose.
 3. **Cross-lineage regulatory swap.** Graft bacterial ACT domains onto mammalian PHGDH (and remove them from bacterial PGDH) to test whether V-type serine feedback is transferable and what constrains its evolution.
 4. **Tissue-resolved flux tracing.** Use stable-isotope (^{13}C -glucose) tracing across mammalian tissues to quantify de novo vs. uptake contributions and map where the module is truly rate-limiting.
 5. **Combination-therapy trials informed by escape routes.** Given documented glutaminolysis escape ([PMID: 40843962](#); [PMID: 41959234](#)), test PHGDH inhibitors combined with glutamine antagonists in serine-auxotroph-prone tumors.
 6. **SerB as an antimicrobial target.** Leverage the *Brucella* and archaeal PSP structures ([PMID: 40719280](#); [PMID: 11342136](#)) to design pathogen-selective phosphatase inhibitors exploiting the phospho-Asp mechanism.
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Report compiled from 8 confirmed findings and 50 reviewed papers across bacterial, archaeal, plant, and mammalian literature. Claims are attributed only to the cited abstracts/snippets; uncertainty is flagged where evidence is indirect or organism-specific.
